Department of Chemistry
CML 100: Organic Chemistry: Home Assignment 1

Instructor: Dr. Nidhi Jain (Room: MS 708)

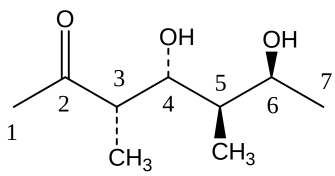
1. The concentration of X dissolved in chloroform is 6.15 g per 100 mL of solution. (a) A portion of this solution in a 5-cm polarimeter tube causes rotation of -1.2°. Calculate the specific rotation of X. (b) Predict the observed rotation if the same solution were placed in a 10-cm tube. (c) Predict the observed rotation if 10 mL of the solution were diluted to 20 mL and placed in a 5 cm tube.

Answer: a) -39.02° b) -2.39° c) α=-0.6°

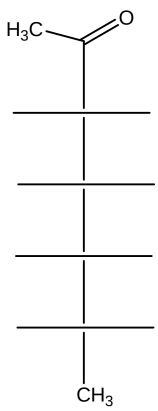
2. The optical rotation of sugar is used in industry as a quick way to check the concentration of sugar solution. If 20 g of cane sugar dissolved in water to make up 100 mL of solution and placed in a tube of 40 cm long rotates the plane of polarized light +53.2° at 20 °C. What is the concentration of another solution of sugar, measured at the same temperature and in the same polarimeter tube, if the optical rotation is +13.3°.

Answer: 0.05 g/mL

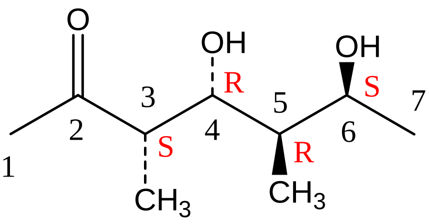
3. Assign absolute configuration at each of the chiral carbons in the given molecule (**B**).
Convert the structure to Fisher projection in the template provided.



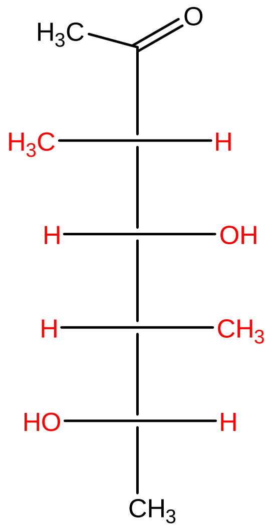
B



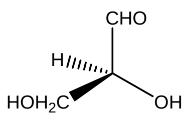
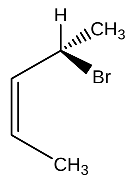
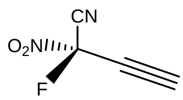
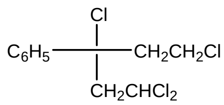
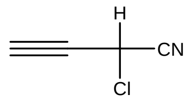
Answer:

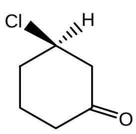
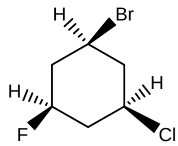
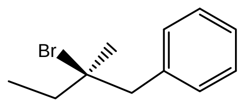
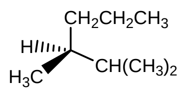
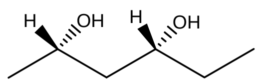


B

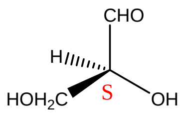


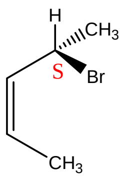
4. Assign R/S configuration to the stereocentres in the following molecules.

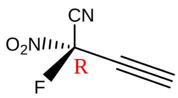
    

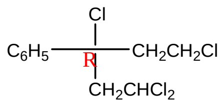
    

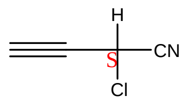
Answer:

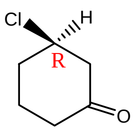


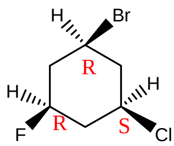


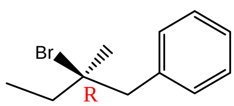


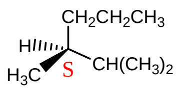


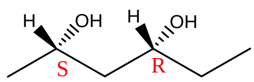




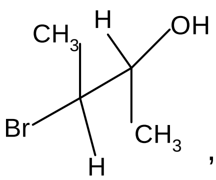
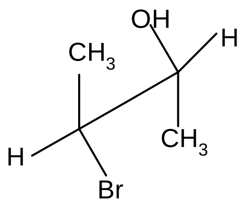
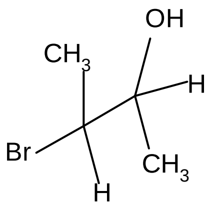
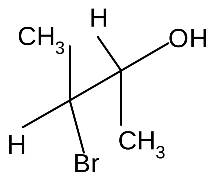




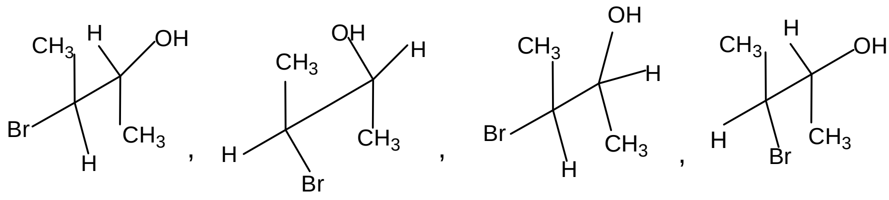




5. Which prefix *threo* or *erythro* is correctly applied to the following structures?

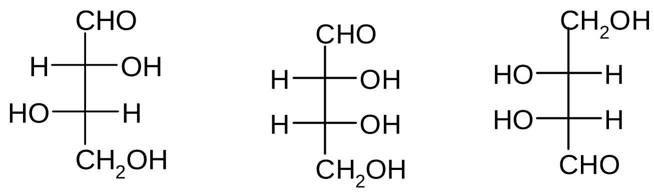
 ,  ,  , 

Answer:



Erythro **Erythro** **Threo** **Threo**

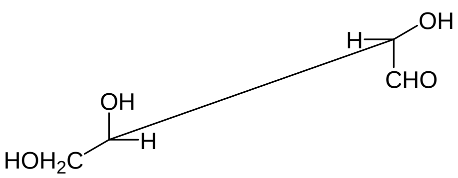
6. Write the Sawhorse and Newman structures for the following compounds and indicate whether any of these structures are superimposable.

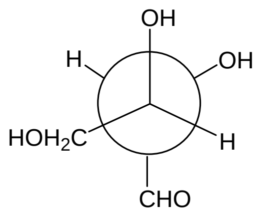


1 2 3

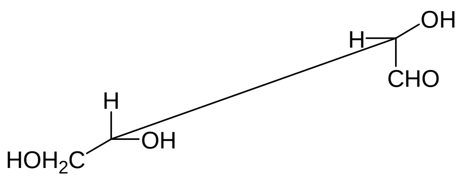
Answer:

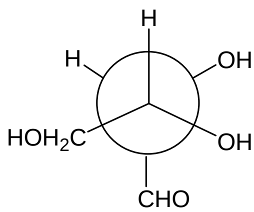
1





2





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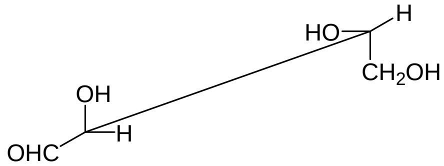
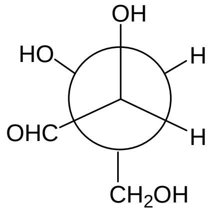
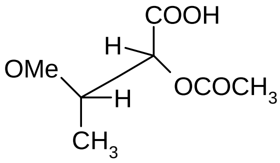
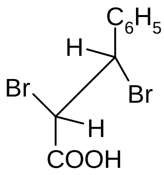




2 and 3 are superimposable

7. Convert the following structure to corresponding Fischer projection formulae.





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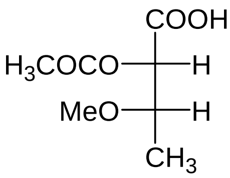
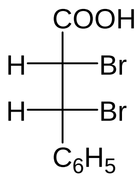
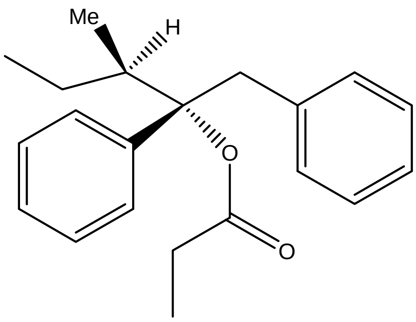


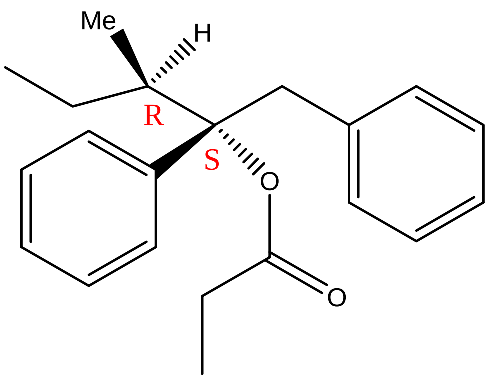


8. Darvon (**1**) is a painkiller. Assign the absolute configurations for the chiral centres if any. Draw the most stable saw-horse and Fischer projections of the molecule.



I

Answer:



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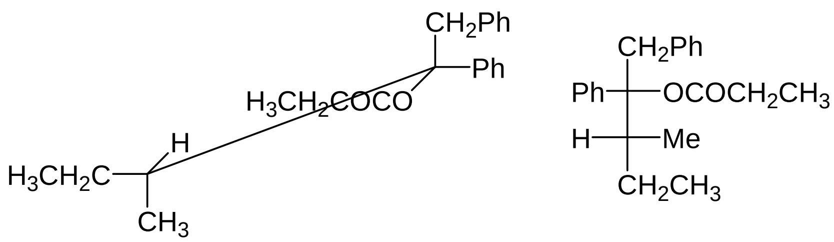
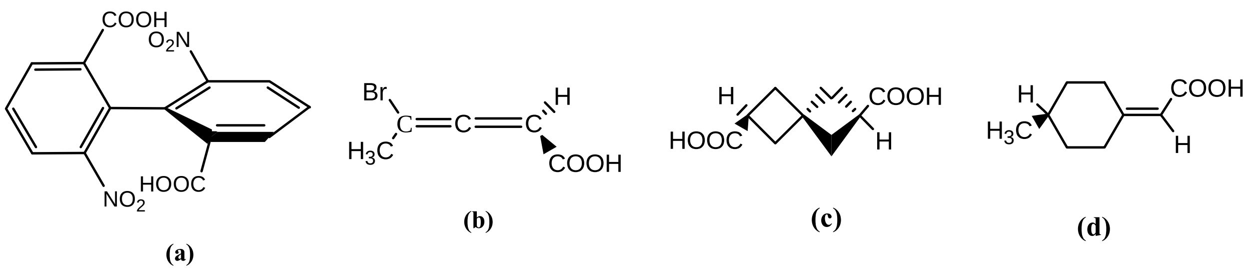
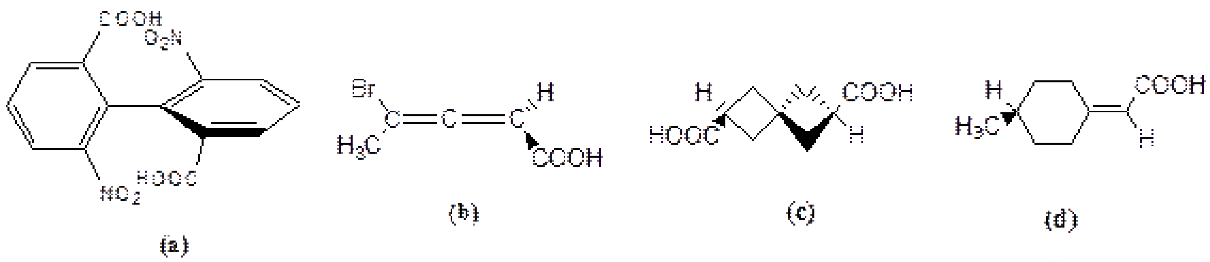


9. Assign absolute configuration to the following compounds.

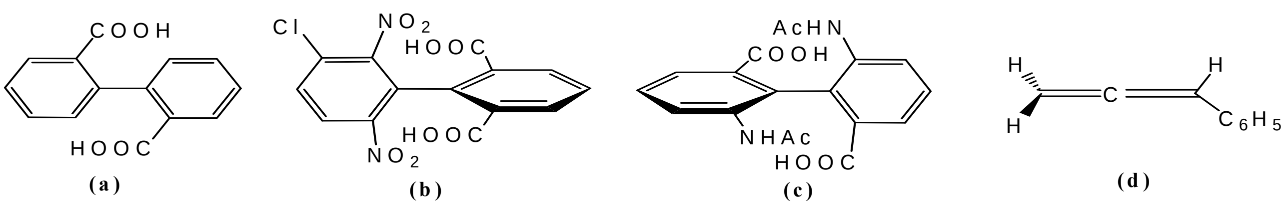


Answer:



S S R R

10. Classify the following molecules as chiral or achiral.



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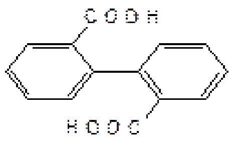
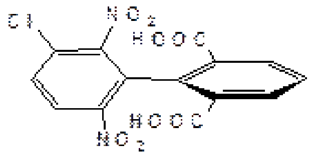


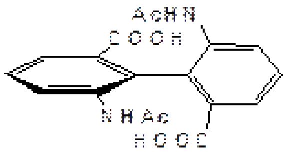
(a)

Chiral



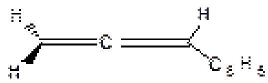
(b)

Achiral



(c)

Chiral



(d)

Achiral